
EDUCATION

Stellenbosch University

Jan 2026 - Present

BSc (Mathematical Sciences) Focal Area: Applied Mathematics

- Majors: Applied Mathematics, Computer Science

PROFESSIONAL EXPERIENCE

Freelance Website Design & Development

Feb 2024 - Present

- Created 10+ websites for small to medium businesses, and owned each project end-to end (communication, design, development, deployment).
- Applied SEO techniques that ranked multiple client sites on the first page of Google for targeted keywords

PROJECTS

AI Budget Coach - Full-stack web app

2026

Next.js, TypeScript, Supabase (PostgreSQL), Google Gemini API

- Built a full-stack budgeting app. Users track income and expenses, visualize spending with interactive charts, and receive AI-generated insights
- Integrated Gemini via a server-side route returning schema-constrained JSON, keeping the key off the client and guaranteeing structured output
- Modelled the data in PostgreSQL with row-level security for per-user isolation. Added email/password auth, insights history, and savings-goal tracking.

Personal Portfolio & CV Site - sebastianmorkel.com

2026

Astro, TypeScript, Cloudflare Pages, Playwright

- Built a statically-generated Astro + TypeScript portfolio on Cloudflare Pages with push-to-deploy CI.
- Authored each project and skill once in a single tagged content collection
- Built a token-based design system with two switchable themes, and automated per-discipline PDF CV generation at build time via headless Chromium (Playwright).

Study Activity Dashboard - Full-stack analytics dashboard

2026

FastAPI, PostgreSQL, Docker, React, TypeScript, TanStack Query, Google Gemini API

- Built a typed FastAPI + PostgreSQL backend (asyncpg). computing per-student stats and a gap-free 14-day activity timeline in SQL with not application code.
- Enforced quiz-score validation (0–100) in two layers; a Postgres CHECK constraint and a Pydantic v2 validator with correct 201/404/422 responses.
- Containerised Postgres with Docker Compose (one-command, seeded startup), and built a React + TypeScript frontend (TanStack Query, Recharts) with an optional Gemini summary

TECHNICAL SKILLS

Languages:

Python, Java, SQL, TypeScript, JavaScript, Visual Basic

Web Technologies:

React, Next.js, Astro, FastAPI, Tailwind CSS, TanStack Query, HTML, CSS, REST APIs

Databases & AI:

PostgreSQL (asyncpg), Supabase (auth, row-level security), Google Gemini API (LLM integration)

Tools:

Git/GitHub, Docker, Cloudflare Pages, Vite, Playwright, VSCode, MS Access, MS Excel